

5. Project Development Phase

Project Module: Visitor Management System: -

Phase Description:

Technical implementation of the Visitor Management System including custom application development, table configuration, client-side validation, Flow Designer automation, email notification setup, and UI Action development.

This document serves as the formal milestone artifact for the ServiceNow development workflow lifecycle.

Application Development & Table Configuration: -

Custom Application Creation:

Created a scoped application named:

Visitor Management System:

within the ServiceNow Personal Developer Instance (PDI).

Visitor Table Creation:

Developed a custom table named:

Visitor:

to manage visitor registration and tracking.

Table Configuration:

Configured the following fields:

Field Name	Type
Number	Auto Number
Visitor Name	String
Employee Name Reference (User)	
Phone Number	String

Field Name	Type
Email	String
Company Name	String
Visit Date	Date
Purpose of Visit	Choice
Status	Choice
Checked In Time	Date/Time

Auto Number Generation:

Configured automatic visitor number generation.

Example:

VIS0001001

VIS0001002

VIS0001003

Choice Field Configuration: -

Purpose of Visit:

Configured the following options:

- Meeting
- Interview
- Delivery
- Vendor Visit
- Other

Status:

Configured the following options:

- Requested
- Approved

- Checked In
- Completed

Default Value

Requested

Client Script Development: -

Phone Number Validation Script:

Developed a client Script to validate visitor phone numbers before record submission.

Validation Logic:

- Accept only 10-digit phone numbers.
- Display an error message if the value is invalid.
- Prevent incorrect data from being saved.

Validation Message:

"Phone Number must contain exactly 10 digits."

Development Outcome:

Ensured improved data quality and reduced invalid visitor registrations.

Flow Designer Development: -

Visitor Registration Flow:

Created a Flow Designer automation named:

Visitor Registration Flow: -

Trigger Configuration:

Trigger Type:

Record Created

Table:

Visitor

Action 1: Update Record:

Configured an Update Record action to automatically update visitor status.

Workflow:

Requested

↓

Approved

Action 2: Send Email:

Configured Flow Designer's native Send Email action.

Email Configuration: -

Recipient:

Visitor Email Address

Subject:

Visitor Registration Approved

Email Contents:

- Visitor Name
- Visit Date
- Purpose of Visit
- Approval Confirmation Message

Development Outcome:

Automated approval processing and eliminated manual notification handling.

UI Action Development: -

Check In Button:

Developed a custom UI Action named:

Check In

Availability Condition:

The button is displayed only when:

Status = Approved

Script Functionality:

When clicked:

1. Status changes to Checked In.
2. Current timestamp is recorded.
3. Success message is displayed.
4. Record is updated automatically.

Result:

Approved Visitor

↓

Check In Button Clicked

↓

Status = Checked In

↓

Checked In Time Recorded

↓

Success Message Displayed

Email Notification Development: -

Approval Notification:

Configured automated email notifications through Flow Designer.

Features:

- Dynamic data population using Data Pills.
- Visitor-specific email generation.
- Automatic dispatch after approval.

Benefits:

- Faster communication.

- Reduced manual effort.
- Improved visitor experience.

Testing & Verification: -

Visitor Registration Testing:

Verified:

- Record creation.
- Auto number generation.
- Field population.

Client Script Testing:

Verified:

- Invalid phone numbers rejected.
- Valid phone numbers accepted.

Flow Testing:

Verified:

- Flow triggers on record creation.
- Status changes from Requested to Approved.
- Email notification is generated.

UI Action Testing:

Verified:

- Check In button visibility.
- Status updates correctly.
- Check-In Time recorded successfully.

Email Testing:

Verified:

- Email sent successfully.

- Correct visitor details included.
- Proper formatting maintained.

5. PROJECT DEVELOPMENT PHASE – VISITOR MANAGEMENT SYSTEM

PROJECT MODULE:
Visitor Management System

PHASE DESCRIPTION:
Technical implementation of custom application, table configuration, client script validation, Flow Designer automation, email notification, and UI Action for visitor check-in.

1. CUSTOM APPLICATION

Created a scoped application named **Visitor Management System**. All components (table, scripts, flows, UI actions, notifications) are kept within this scope.

2. VISITOR TABLE CONFIGURATION

Column Label	Type	Mandatory	Default Value
Number	Auto Number	Yes	Auto
Visitor Name	String	Yes	-
Employee Name	Reference (User)	Yes	-
Phone Number	String	Yes	-
Email	String	Yes	-
Company Name	String	No	-
Visit Date	Date	Yes	-
Purpose of Visit	Choice	Yes	Meeting (Default)
Status	Choice	Yes	Requested
Checked In Time	Date/Time	No	-

Auto Number format: VIS##### (Example: VIS0001001)

3. CLIENT SCRIPT – PHONE NUMBER VALIDATION

```

1 function onSubmit() {
2   var phone = g_form.getValue('u_phone_number');
3   var pattern = /^\d{10}$/;
4
5   if (phone && !pattern.test(phone)) {
6     g_form.addErrorMessage('Phone Number must contain exactly 10 digits.');
```

Validation Message: Phone Number must contain exactly 10 digits.

4. FLOW DESIGNER – VISITOR REGISTRATION FLOW

5. FLOW ACTION – UPDATE RECORD

Action: Update Record
Record: Visitor [u_visitor]
Set fields: Status = Approved

6. FLOW ACTION – SEND EMAIL

Action: Send Email
To: Trigger -> Email
Subject: Visitor Registration Approved
Body (HTML):

```

<h1>Hello <%= (Visitor Name) %>!</h1>
<p>Your visitor registration for <%= (Visit Date) %>
has been approved.</p>
<p>Purpose of Visit: <%= (Purpose of Visit) %>
Thank you.</p>

```

7. UI ACTION – CHECK IN

UI Action Details: Name: Check In, Table: Visitor [u_visitor], Action name: check_in, Type: All rows

UI Action Script:

```

1 current.u_status = 'Checked In';
2 current.u_checked_in_time = new GlideDateTime();
3 current.update();
4 gs.addInfoMessage('visitor checked in successfully.');
```

Condition: u_status = Approved

Success: Visitor checked in successfully.

8. SAMPLE VISITOR RECORDS

Number	Visitor Name	Employee Name	Visit Date	Purpose of Visit	Status	Checked In Time
VIS0001001	John Doe	John Smith	2024-06-15	Meeting	Checked In	2024-06-15 10:35:22
VIS0001002	Jane Brown	Alice Johnson	2024-06-16	Interview	Approved	(empty)
VIS0001003	Robert Lee	Michael Clark	2024-06-17	Requested	Requested	(empty)

9. EMAIL OUTPUT (SAMPLE)

Visitor Registration Approved

no-reply@company.com
to johndoe@gmail.com

Hello John Doe,

Your visitor registration for 2024-06-15 has been approved.

Purpose: Meeting

Thank you.

10. DEVELOPMENT OUTCOME

- ✓ Custom application and table created successfully.
- ✓ Client script validates phone number (10 digits).
- ✓ Flow Designer automation updates status and sends email.
- ✓ UI Action enables security to check in visitors and capture time.
- ✓ System provides complete visitor lifecycle tracking.

Development Outcome:

The development phase successfully delivered a fully functional Visitor Management System within ServiceNow. The solution integrates custom tables, Client Scripts, Flow Designer automation, Email Notifications, and UI Actions to automate visitor registration, approval processing, and check-in management. The completed system improves operational efficiency, enhances visitor tracking, and reduces manual administrative effort.